Re-analysis Summary Report for Peer-evaluation

Paper ID: **Hurst_EvoHumanBehavior_2017_yypJ**

Re-analyst's ID: **htjm4**

Your task

As a peer-evaluator, for this paper you are asked to judge whether the analyst's applied analytical choices for Task 1 and Task 2 are acceptable. By acceptable, we mean that the analysis pipeline is within the variations that could be considered appropriate by the scientific community in addressing the given analytical task. In addition, for Task 1, we ask you to judge whether the reported conclusion adequately follows from the results of the analysis; whether the co-analyst's self-categorization of the result is adequate. Although, you are not required, it would be appreciated, if you could execute the accompanying analysis code to check any mismatch between the results of the analysis and the reported results.

For both tasks, you should provide your evaluation via this survey: https://forms.gle/EAREaC6JjN2PWFfR6

The re-analyst's task

The re-analyst was instructed to assess the following claim of the original authors:
… those with a faster life strategy report greater levels … of psychopathology (p. 1.)

The corresponding article

Here you can find the original article: https://osf.io/tqmbe/?view_only=c874fcf097644e82a320679d4641163a and the original data: https://osf.io/vhmgc/?view_only=5c6bedb36b2549a88ac137d6c746bcb8

They are provided only as context of the re-analysis.

In Task 1, the analyst was asked to conduct the analysis without any restrictions. Here, the analyst reported the following steps and results of the analysis:
In the present analysis, we aimed to examine the claim: "… those with a faster life strategy report greater levels … of psychopathology".

In a first step, we examined the operationalization of the constructs of interest in the original article and a codebook. Life strategy was operationalized via two measures to provide a "more robust measurement of life-history strategy" (p. 4). According to the authors, Mini-K "has been reported as more superior for assessing behavioral traits of life history strategy than compared to the High-K Strategy Scale that suggests more superiority in physical traits" (p.4). Although it could be interesting to focus on which aspect of life history is a better predictor of psychopathology (e.g., via Bayesian multi-model linear regression), we stick to "robustness" goal as is stated by the original authors and we will examine if the score in both scales is correlated with psychopathology (or not). Mini-K consisted of 20 items scored on a 5-point scale. A positive score indicated a slower life strategy. In an original article, the scores were summed, providing acceptable internal consistency. In a dataset, this variable is coded as MiniK_1 to 20. MiniK_Total represents the pre-computed total score provided by the original authors. High-K consisted of 22 items scored on a 5 points scale. As in the previous case, a positive score indicated a slower life strategy. This variable is coded as HKSS_1 to 22 in a dataset. HKSS_Total represents the pre-computed total score provided by the original

authors.
Psychopathology was operationalized via DSM5 to “measure intensity and frequency of symptomatology across mental health domains of 13 psychiatric diagnoses relevant to Diagnostic and Statistical Manual of Mental Disorders” (p.4). DSM5 consisted of 23 items rated on a 5-point scale. This variable is coded as DSM5_1 to 23 in a dataset. DSM5_Total represents a total score in psychopathology pre-computed by the original authors

We decided to work with the pre-computed score provided by the original authors, but before proceeding further, we examined the internal consistency of the scales of interest. As McDonald´s omega is superior (over Cronbach alpha) with regards to less strict assumptions, we recomputed the internal consistency scores for three measures relevant for this analysis. Analysis of internal consistency indicated acceptable results for all three questionnaires (McDonald´s omega = 0.94 for DSM5; 0.81 for Mini-K and 0.91 for HKSS). Before the main analysis, we examined the descriptive statistics of the main variables (note that pre-computed total scores provided by the original authors have been used as they were familiar with questionnaires more than we are and we decided to build upon their expertise in this aspect).

As indicated by the Shapiro-Wilk test for normality, the assumption of normality (both, univariate and multivariate) has been violated. After the inspection of Q-Q plots, we decided to not omit any outliers, but rather to implement non-parametric statistics and robust sensitivity analysis in consequent analysis.

For the analysis, all N=138 participants have been used (46 - 33% were males; 92 – 67% were females), with a mean age of 34.31 years (SD=14.74; Med=29; Min=16; Max=69 years). As we are not aware of any theoretical or legal obligation concerning the minimum age of participation, we decided not to omit participants under 18 years and work with the whole sample in the main analysis. * For example, participants are allowed to take part in this kind of research if they have 16 or more years in our country when (but acceptance of parents is necessary in some cases). We will proceed further with the assumption that it is not a problem in the country of original research team and every legal obligation has been fulfilled. However, to be sure that our decision will not distort the results we will conduct a sensitivity analysis for the main results where participants below 18 years of age will be omitted.

Paraphrasing the claim of interest, our hypothesis stated that: “… those with a faster life strategy report greater levels … of psychopathology”. The additional research question aims to examine if relationship between psychopatology and life strategy will be found in two different operationalizations of life strategies (Mini-K and HKSS).

Considering the hypothesis and addtional goal (to examine the relation between psychopathology measured by DSM5 and life strategies measured by Mini-K and HKSS in terms of robustness) and rather limited number of participants (N=138), we decided to not use a more complex type of analysis (e.g., Structural equation modeling). Rather, a more rudimentary type of analysis suitable for lower N and violations of basic assumptions like normality has been selected. In particular, we decided to use non-parametric correlation analysis. However, to provide more robust conclusions, we decided to conduct various types of sensitivity analysis (with and without selected covariates; with two types of non-parametric analysis; from a perspective of two approaches to statistical inference; and with and without participants below 18 years).

We will start with the main analysis with the classical frequentist approach. We will interpret the results from the perspective of Neyman Pearson approach. To handle type I error (as the selected claim is only part of multiple results reported in the original study), we will use more stringent alpha levels (< 0.005) as recommended by Benjamin et al. (2018) (although such a strategy can increase type II error, our power analysis indicated that with N=138, we should have 99% power to detect r=0.5; 98% power to detect r=0.4 and approximately 80% power to detect r=0.3 which is - considering the effect reported in the original study - our smallest effect size of interest).

First, we conducted a simple correlation analysis with a non-parametric Spearman correlation coefficient without any additional confounders. The results indicated that the DSM5 score was negatively related to both, Mini-K (rs=-0.59, p<0.001) and HKSS (rs=-0.53, p<0.001) scores. This finding supports our hypothesis that those with a faster life strategy report greater levels of psychopathology as more psychopathology was negatively associated with a slower life strategy (we can remind the reader that a positive score indicated a slower life strategy). Considering null hypothesis significance testing, a null hypothesis can be rejected based on alpha level < 0.005. With regards to the interpretation of the effect size, according to standard Cohen´s criteria, this effect could be considered as big in magnitude. However, as Cohen did not mean to use his benchmarks for interpretation, we can use alternative approaches more suitable for interpretation purposes. When compared to the magnitude of the effect that can be found in the field of individual differences, present effects are above 75 percentile in this research field (see Gignac and Szodorai, 2016). Such magnitude is in line with what can be found in the field of clinical psychology (see Schäfer and Schwarz, 2019). From a practical point of view, the effect is (very) large and potentially powerful in both the short and long run, but with a potential caveat that it could be "gross overestimation" of the real effect (see Funder and Ozer, 2019 for further discussion).

Second, we conducted a sensitivity analysis where age and gender were added as confounders. The results are in line with the previous results as DSM5 score was negatively related to both, Mini-K (rs=-0.47, p<0.001) and HKSS (rs=-0.40, p<0.001) scores even when age and gender were accounted for. As in previous case, we can reject null hypothesis. The effect is smaller in magnitude but it still could be considered as relatively big. Third, we added additional confounders related to demographic information (number of siblings, half-siblings, step-siblings as well as answers to questions: Did you live with mum and dad? Do you have a step-mum? Do you have a step-dad? Did your parent have a partner?). Furthermore, we shifted to an alternative non-parametric correlation coefficient (Kendall´s tau) as p-values are more accurate with a smaller sample size and results are in general less sensitive to outliers and errors. The results were in line with the previous analysis as DSM5 score was negatively related to both, Mini-K (rs=-0.33, p<0.001) and HKSS (rs=-0.27, p<0.001) scores. Fourth, we conducted a sensitivity analysis with Bayesian analysis as it quantifies the evidence the data provide for H1 versus H0. For this purpose, the default prior proposed by Jeffreys has been used. Moreover, as in the previous case, Kendall´s Tau correlation was selected to provide more conservative estimate. The results were in line with the previous analysis as the DSM5 score was negatively related to both, Mini-K (rtau=-0.43, BF10=1.51e+11) and HKSS (rtau=-0.37, BF10=1.38e+8) scores. This means that observed data are much more likely under H1 than under H0 – this is "extreme evidence for H1" when word categories are preferred. Last but not least, we conducted a sensitivity analysis of a main results, where only persons older than 18 years are included. The results are in line with the previous analysis as DSM5 score was negatively related to both, Mini-K (rs=-0.54, p<0.001) and HKSS (rs=-0.47, p<0.001) scores.

In sum, when pre-computed total scores are used, more rudimentary analytic strategy in terms of correlation analysis is implemented, non-parametric inference statistic is selected, and when two operationalizations of life strategies are analysed separately, we found a support for the claim that "… those with a faster life strategy report greater levels … of psychopathology" (p. 1.). This was true irrespective of operationalization of the measures for a life strategy (Mini-K or HKSS), inferential school (frequentis vs. Bayesian), inclusion or exclusion of selected confounders (none vs. age and gender vs. various demographic confounders), and the type of analysis (Spearman vs. Kendall´s tau correlation coefficient).

Note that analysis script in JASP will be uploaded to OSF and can be consulted if needed.
Note that as we don´t have expertise regarding the measurements used in original study, we sticked to pre-computed scores available in the dataset. However, this could be limitation as it is possible that different scoring strategies could exists (and be preferred) for DSM5, Mini-K or HKSS. *Note more advanced analysis strategies as Structural equation modelling could be used. SEM could be beneficial for several reasons. For example, measurement error will be taken into account and instead of separate analysis of two measures, latent factor of life strategy could be employed. In particular, SEM could be computed in Lavaan (also implemented in JASP) with Mini-K and HKSS as a first order factors and life strategy as a second order factor or two measures could be used as two latent factors that covary. Furthermore, polychoric correlation matrix and DWLS estimator with robust correction could be used (due to likert-scale items and deviations from a multivariate normality). However, due to small number of observations, we did not find it reasonable to implement such analytic strategy and we implemented more rudimentary correlation analysis.

The analyst's conclusion: We found a support for the claim: "… those with a faster life strategy report greater levels … of psychopathology".

The analyst's categorization of the result:

The results show evidence for the relationship/effect as described in the claim provided in your task

In Task 2, the analysts received the following instruction to conduct the analysis:
You should report the Mini-K measure instead of HKSS for life history strategy.

Here, the analyst reported the following steps and results of the analysis:
In the second task, we aimed to examine the claim: "… those with a faster life strategy report greater levels … of psychopathology".

As in the first task, we examined the operationalization of the constructs of interest in the original article and a codebook at first. Life strategy was operationalized via two measures to provide a "more robust measurement of life-history strategy" (p. 4). However, Mini-K was requested for the second task. Mini-K consisted of 20 items scored on a 5-point scale. A positive score indicated a slower life strategy. In an original article, the scores were summed, providing acceptable internal consistency. In a dataset, this variable is coded as MiniK_1 to 20. MiniK_Total represents the pre-computed total score provided by the original authors. As in a first task, psychopathology was operationalized via DSM5 to "measure intensity and frequency of symptomatology across mental health domains of 13 psychiatric diagnoses relevant to Diagnostic and Statistical Manual of Mental Disorders" (p.4). DSM5 consisted of 23 items rated on a 5-point scale. This variable is coded as DSM5_1 to 23 in a dataset. DSM5_Total represents a total score in psychopathology pre-computed by the original authors

As in the first task, we decided to work with the pre-computed score provided by the original authors, but before proceeding further, we examined the internal consistency of the scales of interest. As McDonald´s omega is superior (over Cronbach alpha) with regards to less strict assumptions, we recomputed the internal consistency scores for three measures relevant for this analysis. Analysis of internal consistency indicated acceptable results for all three questionnaires (McDonald´s omega = 0.94 for DSM5; and 0.81 for Mini-K).

For the analysis correlation analysis have been used as an analytic procedure. The reason why correlation analysis have been selected is that we think it is the most straightforward answer to research question of interest and for corroboration of the main hypothesis: H: "… those with a faster life strategy report greater levels … of psychopathology".

Before the main analysis, we examined the descriptive statistics . As indicated by the Shapiro-Wilk test for normality, the assumption of normality (both, univariate and multivariate) has been violated. After the inspection of Q-Q plots, we decided to not omit any outliers, but rather to implement non-parametric statistics − Kendall´s tau.

For the analysis, all N=138 participants have been used (46 - 33% were males; 92 − 67% were females), with a mean age of 34.31 years (SD=14.74; Med=29; Min=16; Max=69 years). As we are not aware of any theoretical or legal obligation concerning the minimum age of participation, we decided not to omit participants under 18 years and work with the whole sample in the main analysis as in the first task.

For inference, we used both, the frequentist and Bayesian approaches. To handle type I error (as the selected claim is only part of multiple results reported in the original study), we used more stringent alpha levels (< 0.005) as recommended by Benjamin et al. (2018). For BF10 >10 criterion has been used.

DSM5 score was negatively correlated to Mini-K scores (rtau=-0.43, $p<0.001$). Bayesian analysis (with default Stretched beta prior width = 1) indicated evidence for H+; specifically, BF10 = 1.51e+11, which means that the data are more than 1e+11 times more likely to occur under H+ than under H0, indicating extreme evidence in favor of H+ (non directional hypotheses has been used in analysis).

Analysis language/software/code (optional to check):
Task 1: JASP
Task 2: JASP
Analysis codes: https://osf.io/xnwpg/